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AI-based Online Exam Protoring System

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Abstract

This research paper is presented to introduce an AI-based online examination proctoring system which is designed to tackle the alarming increase in the challenges of maintaining the integrity of remote examinations. This system is equipped with advanced AI technologies, including YoLov8 to detect objects, Media pipe to detect the gaze of eyes, and pyaudio for audio analysis, to detect and protect from cheating practices like: unauthorized tab switching, the use of restricted objects(example: smartphones, books) and the presence of several other faces or objects in the exam environment. using Django Framework with PostgreSQL to manage database, this system provides a user-friendly interface for learners, students, and administrators, ensuring the easy registration, login, and real-time examination monitoring. Suspicious activities like gaze diversion, unfamiliar sounds, object detection, several person detections, and tab switching are logged and reported through an admin providing a smooth user experience. Future enhancements that aim to improve accuracy, scalability, and compatibility among devices and to introduce more features like keystroke dynamics analysis. This project remarks the potential of AI so as to create a secure, fair, and effective online exam environment, which helps to make this system a useful tool for educational institutions worldwide.

Keywords

AI-Based Proctoring, Browser Tab Monitoring, Object Detection, YOLOv8, Gaze Tracking, Mediapipe, Audio Analysis, Remote Examination, AI in Education, Academic Integrity, Secure Assessments

1. Introduction

In recent years, online learning and remote education have increased dramatically [1]. Online education platforms have changed the way education is delivered, making it more accessible and flexible for students worldwide. As this trend continues to rise, online examinations have become essential for assessing students' knowledge and abilities. While remote testing offers convenience, it also presents significant challenges in ensuring academic integrity. Traditional methods of exam supervision, which rely on human proctors, suffer from limitations in scalability, consistency, and real-time detection of cheating behavior.

Manual online proctoring is particularly challenging because multiple candidates cannot be monitored simultaneously. During in-person examinations, an invigilator can detect suspicious behavior using visual and auditory cues. However, in online examinations, invigilators are not physically present, making it difficult to detect prohibited actions, such as unauthorized collaboration, tab switching, or the use of external devices. Therefore, an efficient online proctoring system should incorporate AI-based solutions for movement, gaze, and sound detection [2].

To address these challenges, we have developed an AI-based online exam proctoring system. This innovative solution leverages advanced machine learning and computer vision techniques to monitor students' behaviors in real-time, ensuring a secure examination environment. The system is built using the Django framework for the backend and PostgreSQL for database management, ensuring efficiency and scalability.

Unlike existing solutions, our system integrates multiple AI models to enhance cheating detection while minimizing distractions for students. The dataset used for model training consists of 3774 images, with an 80-10-10 split for training, validation, and testing, respectively. We employ the YOLOv8 model due to its superior real-time object detection capabilities, comparing its performance with Faster R-CNN to highlight speed and accuracy differences. However, factors such as varying webcam resolutions, lighting conditions, and internet bandwidth must be considered when deploying the system at scale.

The system's core features include:

- **Browser Tab Switching Detection:** Uses JavaScript to monitor candidates' screen activity, flagging any attempts to switch tabs or access unauthorized resources during the exam.
- **Object Detection:** Employs the YOLOv8 model to detect prohibited objects, such as mobile phones and textbooks, while also identifying multiple individuals in the frame.
- **Gaze Detection:** Utilizes the Mediapipe library to track eye movements, ensuring candidates remain focused on their exam screens throughout the assessment.
- **Audio Analysis:** Uses the PyAudio library to analyze background sounds, detecting unauthorized conversations or noise disturbances. All collected data is securely stored in PostgreSQL for future review.

By integrating these features into a cohesive platform, our AI-based online exam proctoring system offers a scalable and efficient solution for maintaining the integrity of online assessments. The project aims not only to enhance the reliability of remote examinations but also to provide a fair and non-intrusive testing environment for all students.

2. Problem Statement

The growth of online learning platforms has made remote examinations a common practice, but ensuring the integrity of these exams remains a major challenge. While several proctoring applications and systems are available in the market, many of them lag in tackling the effectiveness and user experience and have several critical limitations.

Existing proctoring systems on the market often do not utilize browser tab-switching detection, an important feature to restrict candidates and students from accessing unauthorized resources during exams. Moreover, most applications require all AI modules (e.g., object detection, gaze tracking) to run on the front end, with the webcam feed prominently displayed on the exam page. This can distract students from exams and create unwanted pressure, negatively affecting their performance. Furthermore, many systems produce audible beeps or alerts when suspicious activities are detected, which can frustrate students and disrupt their concentration.

To solve these issues, we present an AI-based online exam proctoring system that solves the limitations of existing solutions. In this system, all AI modules, including object detection, gaze tracking, and audio analysis, run effectively in the background. The webcam feed is also processed in the background, eliminating distractions for candidates. The system provides browser tab-switching detection using JavaScript to monitor and flag unauthorized attempts to access external resources. For audio analysis and detection, the system uses the PyAudio library to record and analyze ambient and background sounds. Unlike traditional and existing systems that produce audible alerts, this project saves all suspicious audio data in the database and produces a detailed report for administrators. This approach ensures a comfortable and student-friendly exam environment.

By overcoming the issues of existing test applications, this project aims to provide a smooth, efficient, and user-friendly solution to maintain the integrity and fairness of online tests.

3. Literature Review

The development of AI-based proctoring systems has gained a lot of attention in recent years, driven by the increasing need for secure and scalable solutions in online examinations. Several studies and technologies have explored the use of AI to monitor and ensure the integrity of online exams. Some institutions require instructor control of online exams to demonstrate and maintain academic integrity, which requires costly monitoring [3]. Costs for students may include the fees of the exam center, the cost of purchasing a remote invigilator, and the time it takes to detect an approved invigilator and manage the exam time. Institutional prices include, for example, staff salaries responsible for managing the examining process, approval of invigilators, maintenance of exam centers, and potential loss of registrations and revenue because not all institutions require invigilators to be online and monitor.

This project examines the possible issues related to online exams and debates that the total costs (student and institutional time and money) outweigh the potential benefits of exam invigilators. To promote academic fairness, the authors present a less expensive, non-protestant alternative, using eight control methods that proctors can use to reduce the likelihood of student cheating. Online education continues to grow, offering opportunities and challenges for students and teachers [4].

One challenge is the perception that academic integrity related to online exams is undermined by unmonitored cheating that results in artificially higher scores. To address these problems, security software has been developed to handle and prevent academic dishonesty. The purpose of this study was to compare the results of supervised and unsupervised online tests. This section reviews relevant work in the field, focusing on browser monitoring, object detection, gaze tracking, and audio analysis.

3.1 Browser Monitoring and Tab Switching Detection

Browser monitoring is a critical component of online proctoring systems. Researchers have explored various methods to detect unauthorized activities, such as tab switching or accessing external resources. For example, a JavaScript-based solution was proposed to track user tab shift activity during online exams, detecting suspicious behavior in real time[5]. Similarly, we developed a browser extension to monitor tab switches and prevent cheating. These approaches highlight the importance of client-side scripting to maintain integrity.

3.2 Object Detection in Proctoring Systems

Object detection has been widely used in detecting systems to identify prohibited objects or unauthorized individuals and to detect multiple people. The YOLO (You Only Look Once) family of models, particularly YOLOv8, has gained popularity due to its real-time detection capabilities and high accuracy. Studies such as this this demonstrated the effectiveness of YOLO-based models in detecting objects such as cell phones, books, and additional persons in exam environments[6]. These results underscore the capability of pre-trained models like YOLOv8for improving proctoring systems.

3.3 Gaze Detection and Eye Tracking

Gaze detection is another key area of research in online proctoring. Mediapipe, an open-source framework, has been widely accepted for real-time gaze tracking due to its seamless capability and ease of integration. Research by utilizing Mediapipe to monitor eye movements and ensure candidates remain focused on their screens[7]. Similarly, a gaze-based proctoring system that detects distractions and alerts invigilators was proposed[8]. These studies demonstrate the effectiveness of gaze detection in maintaining exam integrity.

3.4 Audio Analysis for Suspicious Activity Detection

Audio analysis plays an important role in detecting suspicious and unusual sounds during online exams. Techniques such as noise classification and speech detection have been employed to monitor exam environments. For example, an audio-based proctoring system that detects conversations or unusual noises, highlighting potential cheating attempts was developed [9]. Moreover, it explored the adoption of machine learning algorithms to analyze audio data and identify faults. These approaches emphasize the importance of audio analysis in creating a secure exam environment.

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4. Methodology

4.1 System Architecture

The online proctoring system is designed with a modular architecture that integrates AI-driven monitoring, secure browsing, and real-time data processing, which is given in the following figure.

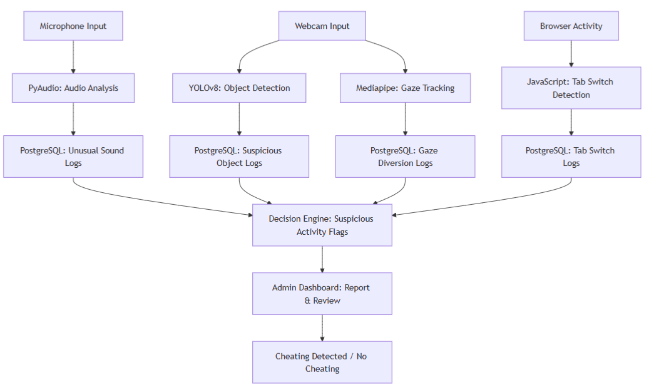


Figure 1: Overall System Architecture

4.2 Technologies and Tools Used

- **Python**: A versatile programming language used for backend development, AI/ML model implementation, and data processing.
- **Django**: A high-level Python framework for building secure, scalable web applications and handling database interactions.
- **JavaScript**: A programming language used for building interactive front-end components and handling real-time client-side functionality.
- **OpenCV**: A computer vision library for processing and analyzing video feeds, enabling tasks like eye-tracking and movement detection.
- **YOLOv8**: A real-time object detection model used to identify unauthorized individuals or devices in the exam environment.
- **MediaPipe**: A cross-platform framework for building pipelines to process images, audio, and video, widely used for facial recognition, hand tracking, and other real-time computer vision tasks.
- **PostgreSQL**: Relational databases used for securely storing user data, exam logs, and performance reports.
- **PyAudio**: A Python library used for working with audio input and output, enabling real-time sound processing for features like microphone access and background noise detection.
- **Face Recognition**: A Python library used for working with facial input and analysis, enabling real-time processing for features like face detection, identity verification, and facial landmark extraction.

4.3 Dataset Collection and Preprocessing

To develop a robust object detection model for online proctoring, we collected a dataset consisting of images captured using OpenCV from live webcam feeds. The dataset includes images of smartphones and books in varied lighting conditions, angles, and positions to ensure diverse representation.

Initially, we manually collected and labeled 900 images (450 for smartphones, 450 for books). To enhance the dataset and improve generalization, we applied augmentation techniques such as rotation, flipping, scaling, and brightness adjustments, expanding the dataset to 3774 images. A detailed breakdown of the dataset is provided in Table 1.

Table 1: Summary of Dataset

Category	Original Images	Augmented Images	Total
Smart phone	450	1437	1887
Book	450	1437	1887
Total	**900**	**2874**	**3774**

4.4 Data Annotation and Labeling

Accurate annotation is crucial for training an object detection model. We used the Roboflow platform to manually annotate each image by drawing bounding boxes around objects. Each image was carefully labeled to distinguish between "smartphone" and "book" classes. Ensuring high-quality annotations minimizes false positives during object detection.

4.5 Data Splitting

To evaluate model performance, we divided the dataset into training, validation, and test sets using an 80-10-10 split. The distribution of images across these sets is shown in Table 2.

Table 2: Dataset Split Distribution

Set	Percentage	Images
Training	80%	3019
Validation	10%	377
Test	10%	378
Total	100%	3774

The training set was used to optimize model parameters, the validation set guided hyperparameter tuning, and the test set provided an unbiased estimate of model performance. This structured division ensures better generalization and robustness.

4.6 Data Augmentation

To address dataset limitations and improve generalization, we employed data augmentation. We applied transformations such as:

- **Geometric Transformations**: Rotation, flipping, and cropping to simulate varied object orientations.
- **Photometric Adjustments**: Brightness and contrast alterations to account for different lighting conditions.
- **Scaling and Translation**: Ensuring model robustness to varying object sizes and positions.

Augmentations were applied using Roboflow, which generated additional training samples while preserving dataset diversity.

4.7 AI Models and Algorithms

The following are the model and algorithm that we have used for this project.

4.7.1 Object Detection Model

YOLO model
YOLOv8n(YOLOv8 Nano) is a lightweight version of the YOLOv8 object detection model. Since YOLO models are pre-trained on the COCO dataset, which includes 80 object classes such as persons, smartphones, books, laptops, cats, and more, we were able to leverage this existing knowledge. As these classes already encompass common objects like smartphones and books, we only needed to fine-tune the model to detect cheating tools in exam environments better.

The Yolo Architecture [10] can be seen in Fig 2.

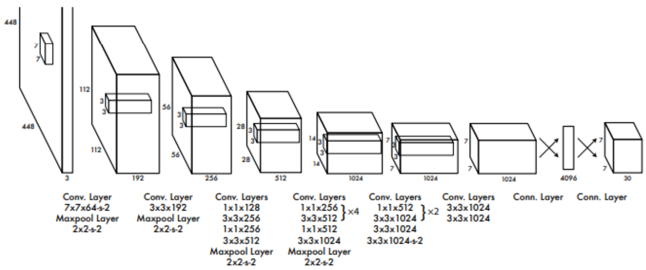


Figure 2: Yolo Architecture

Faster R-CNN Model
Faster R-CNN (Region-based Convolutional Neural Network) is a two-stage object detection model that improves upon Fast R-CNN by introducing a Region Proposal Network (RPN). This RPN eliminates the need for external region proposal methods, making the model faster and more efficient for object detection.

In our proctoring system, Faster R-CNN is used to detect cheating tools such as smartphones and books with higher accuracy. Unlike YOLO, which performs detection in a single step, Faster R-CNN first generates region proposals and then classifies objects within those regions. This two-stage approach allows for more precise object localization but at the cost of increased computational time. The Faster R-CNN Architecture [11] can be seen in Fig 3.

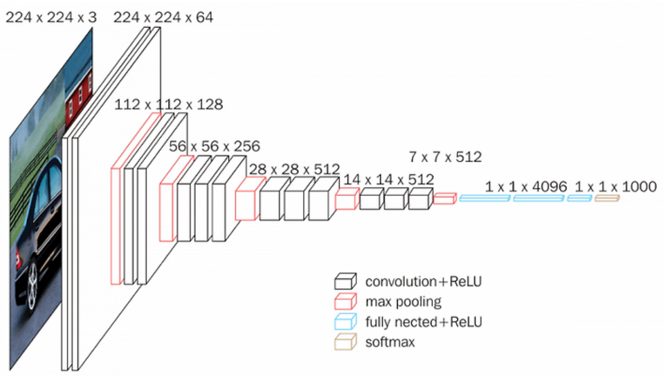


Figure 3: Faster R-CNN Architecture

We have chosen the YOLOv8n model for object detection in this project because of its high speed and accuracy in real time. To further improve the model's accuracy, we fine-tuned it using our custom-collected dataset, which specifically contains images of smartphones and books. This customization allowed the model to become more adept at detecting these objects in the context of online exams.

After training, the model's performance was evaluated using standard metrics such as mean average precision (mAP), intersection overunion (IoU), precision, recall, and inference speed.

4.7.2 Transfer Learning for Object Detection

Transfer learning is a deep learning technique where a pre-trained model is used for a new task by fine-tuning its parameters. The model is not trained from scratch, but it is trained to make some layers of the neural network freeze and only trained the last layer with the custom dataset.

In our AI-based online proctoring system, we fine-tune the YOLO model to detect objects such as mobile phones and books. The training process involves modifying the final layers of the pre-trained YOLO model to adapt to the new dataset.

Mathematical Formulation of Transfer Learning

The training process updates the model parameters θ iteratively to minimize the total loss function $\mathscr{L}_{\text{total}}$. The weight update rule using gradient descent is given by:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \mathscr{L}_{\text{total}} \tag{1}$$

where:

- θ represents the trainable parameters of the model.
- t denotes the training iteration step.
- η is the learning rate, which controls the step size of parameter updates.
- $\nabla \mathscr{L}_{\text{total}}$ is the gradient of the total loss function with respect to θ.

YOLO Loss Function

The total loss function for YOLO, which consists of localization loss, confidence loss, and classification loss, is formulated as[?]:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{coord}}\mathcal{L}_{\text{coord}} + \mathcal{L}_{\text{conf}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} \tag{2}$$

where:

- $\mathcal{L}_{\text{coord}}$ - Localization loss for bounding box regression.
- $\mathcal{L}_{\text{conf}}$ - Confidence loss for object presence.
- $\mathcal{L}_{\text{cls}}$ - Classification loss for object categories.
- λ_{coord} and λ_{cls} are weighting factors.

Weight Update using Adam Optimizer

For better convergence, we use the Adam optimizer, which updates weights using adaptive moment estimation[12]:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\nabla\mathcal{L}_{\text{total}} \tag{3}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)(\nabla\mathcal{L}_{\text{total}})^2 \tag{4}$$

$$\theta^{(t+1)}_{\text{trainable}} = \theta^{(t)}_{\text{trainable}} - \frac{\eta}{\sqrt{v_t}+\epsilon} m_t \tag{5}$$

where:

- m_t and v_t are the first and second moment estimates.
- β_1 and β_2 are exponential decay rates.
- ϵ is a small constant to prevent division by zero.

4.7.3 Face Recognition

Face recognition is a biometric technology that identifies or verifies individuals based on facial features. It is widely used in security systems, identity verification, and human-computer interaction.

In this project, the user's face is recognized and converted into a numerical face encoding, which is then stored in the database. During login, the user's face is captured again, and its encoding is generated and compared with the stored encoding. If the similarity between the two encodings is within an acceptable threshold, the user is authenticated successfully.

The mathematical equation:

Given an input face image I, a feature extraction function $f(I)$ maps it to a high-dimensional feature vector X:

$$X = f(I), \quad X \in \mathbb{R}^n \tag{6}$$

where n is the number of extracted features. To compare a stored face encoding X_1 with a new face encoding X_2, a distance metric $d(X_1, X_2)$ is used:

Similarity Measurement

- **Euclidean Distance**

$$d(X_1, X_2) = \sqrt{\sum_{i=1}^{n}(X_{1i} - X_{2i})^2} \tag{7}$$

  - Measures the straight-line distance between two feature vectors[13].
  - Lower values indicate higher similarity between faces.

- **Cosine Similarity**

$$d(X_1, X_2) = \frac{X_1 \cdot X_2}{\|X_1\|\|X_2\|} \tag{8}$$

  - Measures the cosine of the angle between two feature vectors[14].
  - Values range from -1 to 1, where 1 means identical and 0 means completely different.
  - Used when magnitude differences do not matter.

4.7.4 Face Detection

Face detection is a computer vision of Artificial intelligence that identifies and locates human faces in images or video frames. It involves detecting facial features, such as eyes, nose, and mouth, and drawing bounding boxes around detected faces.

The detection process involves identifying facial landmarks and drawing bounding boxes around detected faces. MediaPipe's model is optimized for both CPU (Central Processing Unit) and GPU (Graphical Processing Unit), which allows it to run efficiently even on low-power devices.

Furthermore, the library's lightweight nature allows seamless integration into web application solutions, making it a robust choice for AI-powered monitoring.

Mathematical Equation

- **Loss Function:**

$$L = \frac{1}{N}\sum_{i=1}^{N} L_{\text{cls}}(c_i, c_i^*) + \lambda \frac{1}{N_{\text{pos}}}\sum_{i=1}^{N_{\text{pos}}} L_{\text{reg}}(B_i, B_i^*), \tag{9}$$

  where:

  - L_{cls} is the classification loss (e.g., cross-entropy loss) that evaluates how well the model distinguishes faces from non-faces[11].
  - L_{reg} is the regression loss (e.g., smooth L1 loss) that measures the accuracy of the predicted bounding box coordinates.
  - c_i is the predicted confidence score for the i-th detection, and c_i^* is the corresponding ground truth label.
  - B_i and B_i^* denote the predicted and ground truth bounding box coordinates, respectively.
  - λ is a balancing factor that weights the regression loss relative to the classification loss.
  - N is the total number of predictions and N_{pos} is the number of positive (face) detections.

4.7.5 Gaze Detection

Gaze detection is a computer vision technique used to track and analyze a person's eye movements, which in turn determines their point of focus[15]. Here, gaze detection is implemented using the Mediapipe Face Mesh, a robust tool for extracting facial landmarks.

Mathematical Equation

Let $p_{33} = (x_{33}, y_{33})$ and $p_{159} = (x_{159}, y_{159})$ denote two key landmarks for the left eye, and $p_{362} = (x_{362}, y_{362})$ and $p_{386} = (x_{386}, y_{386})$ for the right eye. The centers of the left and right eyes, C_L and C_R respectively, are computed as:

$$C_L = \left(\frac{x_{33} + x_{159}}{2}, \frac{y_{33} + y_{159}}{2}\right)$$

$$C_R = \left(\frac{x_{362} + x_{386}}{2}, \frac{y_{362} + y_{386}}{2}\right)$$

Based on these centers, the horizontal gaze direction is determined as follows:

$$\text{Gaze} = \begin{cases} \text{left} & \text{if } C_{L,x} < 0.4, \\ \text{right} & \text{if } C_{R,x} > 0.6, \\ \text{center} & \text{otherwise.} \end{cases}$$

The gaze detection module continuously monitors eye movements in real-time. If a student frequently looks away from the exam screen or maintains an off-screen focus for an extended period, the system flags this as suspicious behavior. By integrating gaze detection with other monitoring techniques, our platform ensures that students remain engaged with their exams, thereby reducing the likelihood of cheating.

4.7.6 Audio Detection

Audio detection is the process of identifying, analyzing, and classifying sounds using various techniques such as signal processing and machine learning. This involves extracting features from input audio signals, ls such as frequency, amplitude, and duration, in order to recognize specific patterns.

PyAudio is a library that works based on digital signal processing (DSP) principles, primarily dealing with sampling, Fourier transforms, and filtering.

Sampling Theorem

$$x[n] = x(nT_s) \tag{10}$$

where:

- $x[n]$ is the discrete-time signal,
- $x(t)$ is the original continuous-time signal,
- $T_s = \frac{1}{f_s}$ is the sampling period,
- f_s is the sampling frequency.

To avoid aliasing, the Nyquist criterion states:

$$f_s \geq 2f_{\max} \tag{11}$$

where $f_{\max}$ is the highest frequency in the signal.

Fourier Transform (FFT)itebhandari2019shift:

To analyze the frequency content of an audio signal, PyAudio applies the Discrete Fourier Transform (DFT), computed using the Fast Fourier Transform (FFT):

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{kn}{N}} \tag{12}$$

where:

- $X(k)$ represents the frequency-domain signal,
- $x(n)$ is the time-domain signal,
- N is the total number of samples,
- j is the imaginary unit.

Windowing Function To reduce spectral leakage, PyAudio applies a window function:

$$x_w(n) = x(n) \cdot w(n) \tag{13}$$

where $w(n)$ is a window function such as the Hamming window:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \tag{14}$$

Digital Filtering (Convolution for Signal Processing) If PyAudio is used for filtering, it applies discrete convolution:

$$y[n] = \sum_{k=0}^{M} h[k] x[n-k] \tag{15}$$

where:

- $y[n]$ is the filtered signal,
- $h[k]$ is the impulse response of the filter,
- $x[n]$ is the input signal.

5. Model Evaluation

To assess the performance of our object detection model in the online proctoring system, we use the following standard evaluation metrics:

5.1 Model Metrics

Precision:

Precision measures the proportion of correctly detected objects among all detections. It is defined as:

$$Precision = \frac{TP}{TP+FP} \tag{16}$$

where TP represents the number of true positives and FP represents the number of false positives.

Recall:

Recall quantifies the ability of the model to detect all relevant objects in an image. It is given by:

$$Recall = \frac{TP}{TP+FN} \tag{17}$$

where FN represents the number of false negatives.

F1-Score:

The F1-score is the harmonic mean of precision and recall, providing a balanced measure of both metrics:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{18}$$

5.2 Comparison of YOLOv8 and Faster R-CNN

Table 3: Comparison of YOLOv8 and Faster R-CNN

Model	Class	Precision	Recall	F1-Score
YOLOv8	Book	0.709	0.912	0.800
	Smart Phone	0.56	0.818	0.664
Average	–	**0.6345**	**0.865**	**0.732**
Faster R-CNN	Book	0.9524	0.9722	0.962
	Smart Phone	0.9800	0.9655	0.9727
Average	–	**0.9662**	**0.9688**	**0.9673**

5.3 Confusion Matrix

The confusion matrix provides a visual representation of the model's classification performance by showing the distribution of true positives, false positives, false negatives, and true negatives. It helps in understanding model errors and improving detection accuracy.

5.4 Validation Curves

Validation curves illustrate how the model's performance evolves during training, providing insights into underfitting or overfitting. These curves help in fine-tuning hyperparameters to improve generalization.

Why YOLOv8 over Faster R-CNN
From the results, Even though Faster R-CNN performs better than YOLOv8 in terms of accuracy, with higher precision, recall, and F1 scores, YOLOv8 is the better choice for the

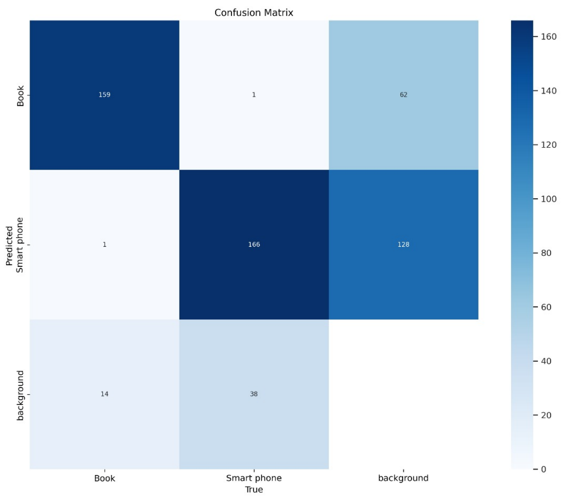


Figure 4: Confusion Matrix for YOLOv8

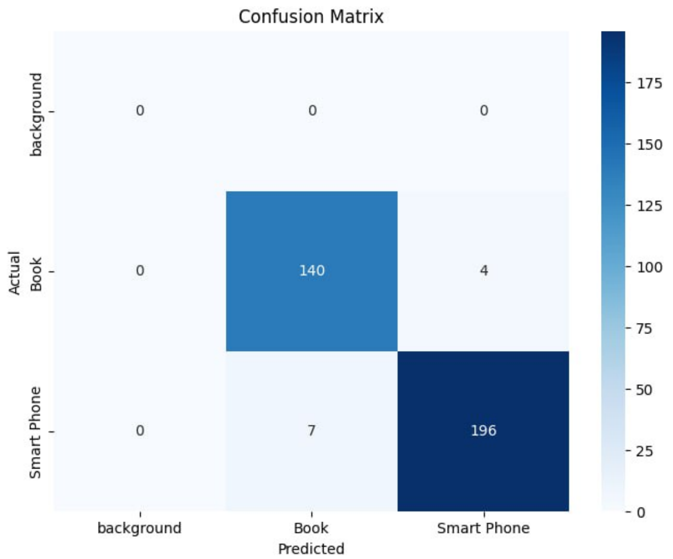


Figure 5: Confusion Matrix for Faster R-CNN

online exam proctoring system. The reason for this is that YOLOv8 is faster and can detect objects in real time, which is crucial for monitoring students during an exam. While YOLOv8's accuracy is a bit lower, its speed makes it ideal for tracking cheating activities during live exams, making it the more practical option for this system.

6. Results and Discussion

The AI-Based Online Exam Proctoring System was developed and rigorously tested to assess its ability to detect and prevent various cheating methods during online exams. It effectively flagged suspicious behaviors—such as having a partner present, using a cell phone, excessive browser tab switching, and abnormal gaze patterns—while logging all activities in real-time.

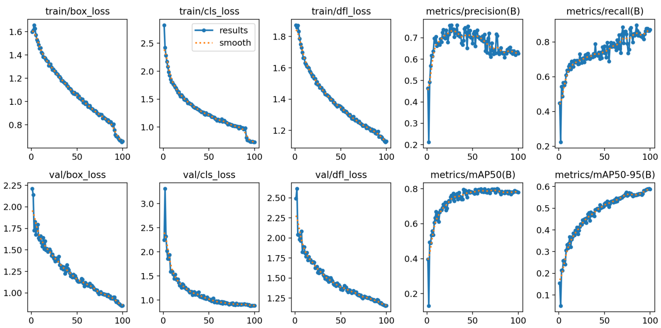


Figure 6: Validation Curve for YOLOv8

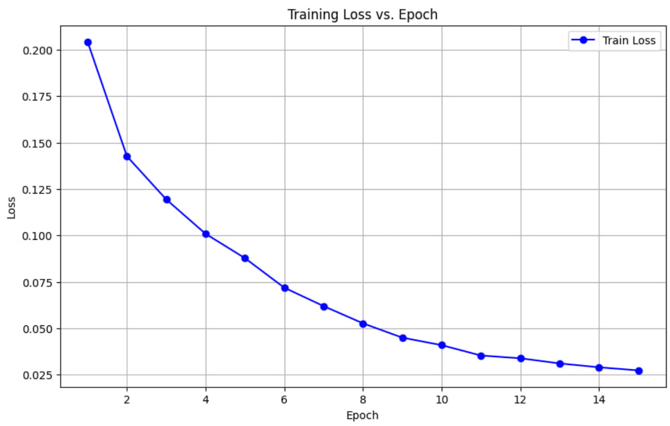


Figure 7: Training Loss curve for Faster R-CNN

6.1 Results

6.1.1 User Interface and Workflow

The system provides a user-friendly interface for both examinees and administrators. Key screens include:

- **Registration Page:** Users enter their details and capture a photo, which is stored in the PostgreSQL database.

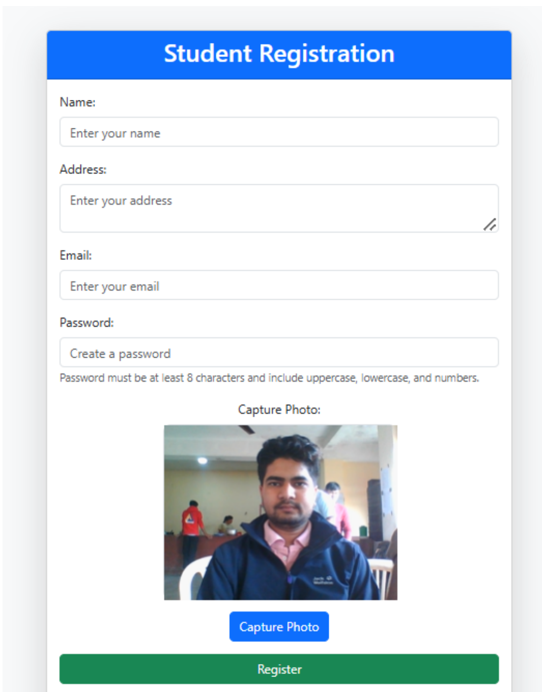


Figure 8: Screenshot of the Registration Page

- **Login Page:** Users log in with their email, password, and a live-captured photo for facial recognition.

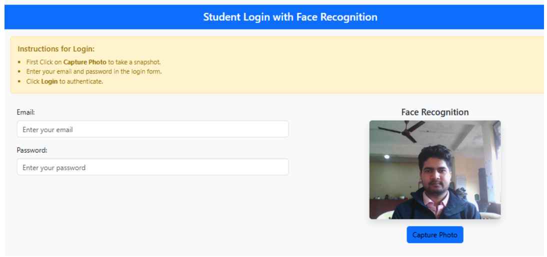


Figure 9: Screenshot of the Login Page

- **Exam Page:** This is the main interface where the exam is conducted and the AI monitoring modules operate in the background.

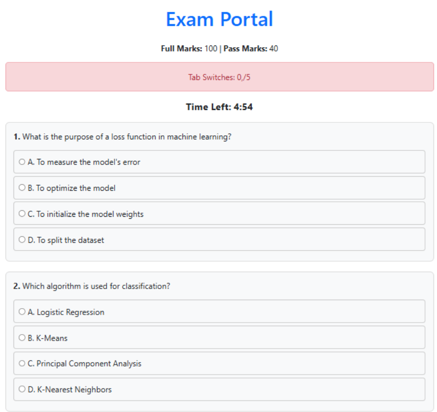
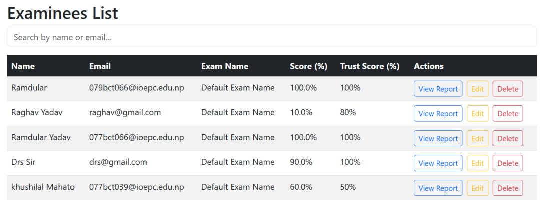


Figure 10: Screenshot of the Exam Interface

- **Admin Dashboard:** Administrators can review detailed reports and logs of suspicious activities.



Examinees List

Search by name or email...

Name	Email	Exam Name	Score (%)	Trust Score (%)	Actions
Ramdular	079bct066@ioepc.edu.np	Default Exam Name	100.0%	100%	View Report Edit Delete
Raghav Yadav	raghav@gmail.com	Default Exam Name	10.0%	80%	View Report Edit Delete
Ramdular Yadav	077bct066@ioepc.edu.np	Default Exam Name	100.0%	100%	View Report Edit Delete
Drs Sir	drs@gmail.com	Default Exam Name	90.0%	100%	View Report Edit Delete
khushilal Mahato	077bct039@ioepc.edu.np	Default Exam Name	60.0%	50%	View Report Edit Delete

Figure 11: Screenshot of the Admin Dashboard

6.1.2 Proctoring Mechanism

The system integrates several AI modules to monitor exam behavior in real-time:

- **Browser Tab Switching Detection:** A JavaScript module tracks tab switches and flags any suspicious activity.

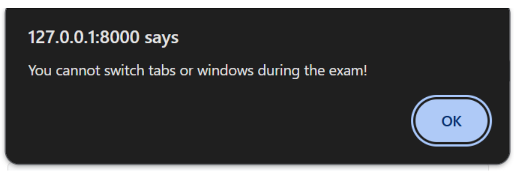


Figure 12: Browser Tab Switching Detection Log

- **Object Detection:** Using the pre-trained YOLOv8 model, the system identifies prohibited items, such as smartphones and books.

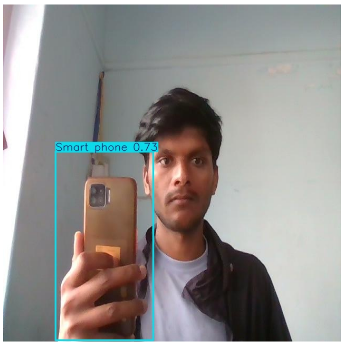


Figure 13: Object Detection Module Output

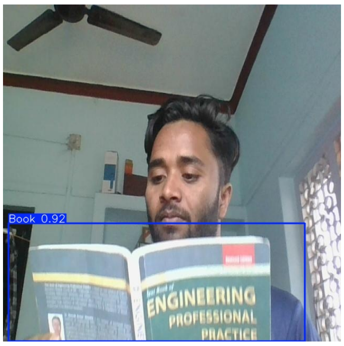


Figure 14: Object Detection Module Output

- **Gaze Detection:** Leveraging the Mediapipe library, this module monitors candidates' eye movements to ensure their focus remains on the exam screen.

Figure 15: Gaze Detection Module Output

- **Audio Analysis:** Ambient sounds are recorded and analyzed to detect unusual noises that may indicate cheating.

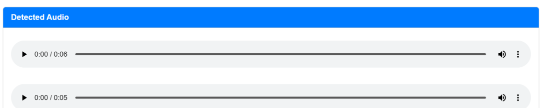


Figure 16: Audio Analysis Module Output

- **Multiple Person Detection:** Using the pre-trained YOLOv8 model, the system detects when more than one person is present in the exam environment.

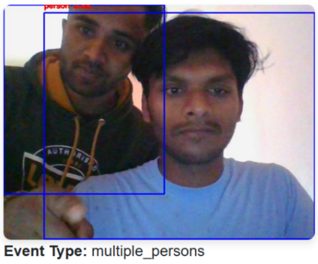


Figure 17: Multiple Person Detection Output

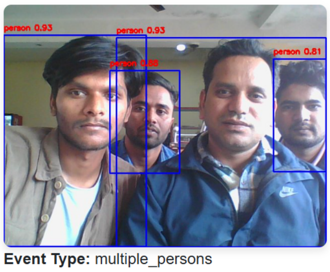


Figure 18: Multiple Person Detection Output

6.1.3 Cheating Summary

At the conclusion, the system displays a summary message that indicates whether any suspicious activity was detected. If any suspicious behaviors (such as unauthorized tab switching, detection of prohibited objects, extra persons, or abnormal gaze patterns) are recorded, a "Cheating Detected" message is shown. Otherwise, a "No Cheating Detected" message is displayed.

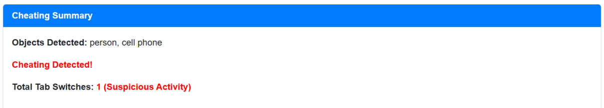


Figure 19: Cheating Detected Message

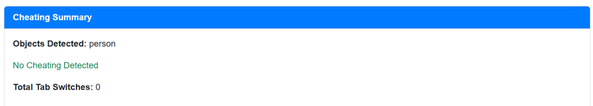


Figure 20: No Cheating Detected Message

6.1.4 Logging and Reporting

All suspicious activities detected during the exam are logged in real-time and stored in the PostgreSQL database. These logs include:

- Timestamps of each tab switch.
- Images captured during object detection events.
- Records of gaze diversion events.

- Audio recordings of unusual sounds.

The admin dashboard provides a consolidated view of these logs, enabling administrators to review the exam session thoroughly.

6.2 Discussion

The evaluation results confirm that this system effectively detects common cheating practices during online exams. The combination of multiple AI modules, each specializing in a different aspect of monitoring, real-time detection, and logging of suspicious activities.

This system is designed with using Django as a Python backend framework and PostgreSQL for storing data, which has provided a stable and secure backend for managing user data and exam logs, while JavaScript offers precise real-time monitoring of browser activity. The integration of pre-trained models, such as YOLOv8 for object detection and Mediapipe for gaze tracking, has allowed us to achieve high detection accuracy.

Including images of each key component (registration, login, exam interface, and detection logs) in this paper serves to illustrate the system's practical implementation and validate its performance. These visuals not only help in understanding the workflow but also enhance the transparency and credibility of our results.

Overall, the AI-Based Online Exam Proctoring System presents a scalable and efficient solution for maintaining exam integrity. The system's modular design and comprehensive monitoring capabilities provide a promising foundation for secure online assessments.

7. Conclusion

The AI-Based Online Exam Proctoring System successfully addresses the critical challenges of maintaining integrity and fairness in online examinations. By integrating advanced AI technologies such as YOLOv8 for object detection, Mediapipe for gaze tracking, and audio analysis modules, the system provides a robust and scalable solution for detecting and preventing cheating practices. These include unauthorized tab switching, the use of prohibited items, the presence of additional individuals, and suspicious audio activities.

Furthermore, built on the Django framework and utilizing PostgreSQL for database management, the system offers a user-friendly and efficient platform for both examinees and administrators. The integration of multiple AI modules ensures real-time monitoring and accurate detection of cheating behaviors, while the admin dashboard provides detailed reports and logs for further analysis.

The system's effectiveness was demonstrated through its ability to detect various cheating practices, such as sitting with a partner, using smartphones, and switching tabs. The implementation of adaptive thresholding and AI-driven decision-making enhances the reliability of cheating detection while minimizing false positives.

Finally, the AI-Based Online Exam Proctoring System marks a significant advancement in online exam security. By integrating cutting-edge AI technologies with an intuitive interface, it offers a reliable and innovative solution to uphold the integrity of remote assessments, bridging the gap between traditional examination methods and AI-powered proctoring systems. AI is transforming the education industry by enabling real-time facial recognition, eye-tracking, and behavior analysis for secure online exams while also enhancing learning through adaptive platforms that personalize content based on student progress.

8. Future Work

While the system has shown promising results, there are several areas for future improvement:

- **Hardware Independence:** Optimize the system to perform effectively across a wider range of hardware configurations, ensuring consistent performance even with low-quality webcams and microphones.
- **Scalability:** Develop strategies to handle large-scale exams with thousands of candidates, ensuring the system remains efficient and responsive under heavy load.
- **Bias Detection and Mitigation:** To ensure fairness, bias mitigation techniques such as adaptive thresholding for diverse skin tones and varying lighting conditions will be explored in future iterations.
- **Cross-Platform Compatibility:** Extend the system's compatibility to mobile devices, ensuring a seamless experience for users on different platforms.
- **Optimized Model Deployment:** Future work will focus on optimizing model deployment for different computing environments, including low-power devices, to enhance accessibility and efficiency.
- **Real-World Deployment Challenges:** Addressing potential real-world constraints such as internet bandwidth limitations, user adaptability, and regulatory compliance to ensure widespread usability.

By addressing these areas, the system can further improve its accuracy, fairness, and usability, making it a more effective solution for secure online examinations.

Acknowledgments

The development of this AI-Based Online Exam Proctoring System would not have been possible without the continuous support, guidance, and encouragement of institutions. We sincerely thank the Department of Electronics and Computer Engineering, IOE Purwanchal Campus, Dharan, for providing the necessary infrastructure, resources, and academic support for this research. Their valuable insights and constructive feedback have played a significant role in shaping this project.

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